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Data hiding in digital images using a partial optimization technique based on classical LSB method

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Abstract

This paper presents a new partial optimization for LSB data hiding approach that can be used for protecting of any secret information or data. A deterioration effect as little as possible in an image is intended using the LSB data hiding technique and, this is well realized utilizing the proposed partial optimization approach achieving the same data embedding bit rates. In the proposed approach, the entire image pixels are classified into eight regions and then it is applied to the eight distinct ordering coding to the each region by the developed partial optimization encoder. Thus, the most effective outcome that has been obtained from the eight regions means that the number of the altered bits is kept minimized. In this way, the minimal values that have been attained from eight regions enable decoding that ensures relatively small distortions on the extracted cover image.

Keywords: *data hiding, steganography, LSB, optimization.*

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1. Introduction

In recent years, techniques for data hiding have become increasingly more sophisticated and widespread. Data hiding is a technique to embed the original secret image to another cover image by some encryption/encoding techniques. Researches on information embedding, particularly information hiding techniques, have received considerable attention within the last years due to its potential application in multimedia and information security [1].

Data hiding has been designed to make it harder for the users to find out the data through hiding it in forms of various materials such as, image, text, audio, and video. The confidential document message to be transmitted is camouflaged in a carrier so that its detection becomes difficult. Information related to the sender and the receiver of the message also can be hidden this way [2]. The embedded data should not to be perceivable to a human sense since it should be invisible and inaudible.

Data hiding is closely connected with cryptology. The purpose of the cryptology is that is transformed of the messages with unintelligible forms and thus if whoever has not the secret key, who is not obtain original message. Sometimes, it is desired that increasing of security and privacy, instead of changing ciphered messages, by means of masking of communication [3]. In data hiding, the message should not be connected with an image; on the contrary it is to be embedded. This communication type can be named as reliable secret communication because the message cannot be obtained any information. The goal of data hiding is to avoid drawing suspicion to the transmission of a hidden message. If suspicion is raised, then this goal is defeated. Discovering and rendering useless such covert messages is a new art form known as attacks [4]. Data

hiding techniques represents miscellaneous utilities such as copyright information (watermarking), critical information hiding, into various forms of communication channel such as text, image, audio and video [5]. The interceptors cannot be notice about the existence of the embedded data in the covered image. It is intended that minimal deterioration in image and maximal storage of secret data. Its purpose is not restricted or regulates access to the host signal, but rather to ensure that embedded data remain inviolate and recoverable. From this point of view, the main objective of the presented research work is to propose a new partial optimization and LSB based data hiding approach for digital images.

Rest of the paper is organized as follows. Section 2 summarizes Least Significant Bits (LSB) data hiding technique. Section 3 describes the new paradigm for the proposed partial optimization based LSB data hiding approach. Section 4 presents the experimental results of the proposed methods and a detailed evaluation study followed by final remarks in the last section.

2. Least Significant Bits (LSB) Data Hiding

The simplest data hiding techniques embed the bits of the message directly into the least significant bit plane of the cover image in a deterministic sequence. Modulating the least significant bit does not result in a human-perceptible difference because the amplitude of the change is too small. Other techniques process the message with a pseudorandom noise sequence before or during insertion into the cover image. The advantage of LSB embedding is its simplicity and many techniques use these methods [6-9]. LSB embedding also allows high perceptual transparency. However, there are many weaknesses when robustness, tamper resistance, and other security issues are

considered. LSB encoding is extremely sensitive to any kind of filtering or manipulation of the covered image. Scaling, rotation, cropping, addition of noise, or lossy compression to the covered image is very likely to destroy the message. Furthermore, if an attacker suspects, he/she can easily remove the message by removing (zeroing) the entire LSB plane with very little change in the perceptual quality of the modified covered image. Specific LSB technique that can embed data in an image, audio and text covers. Figure 1 shows a LSB application example for a RGB pixel.

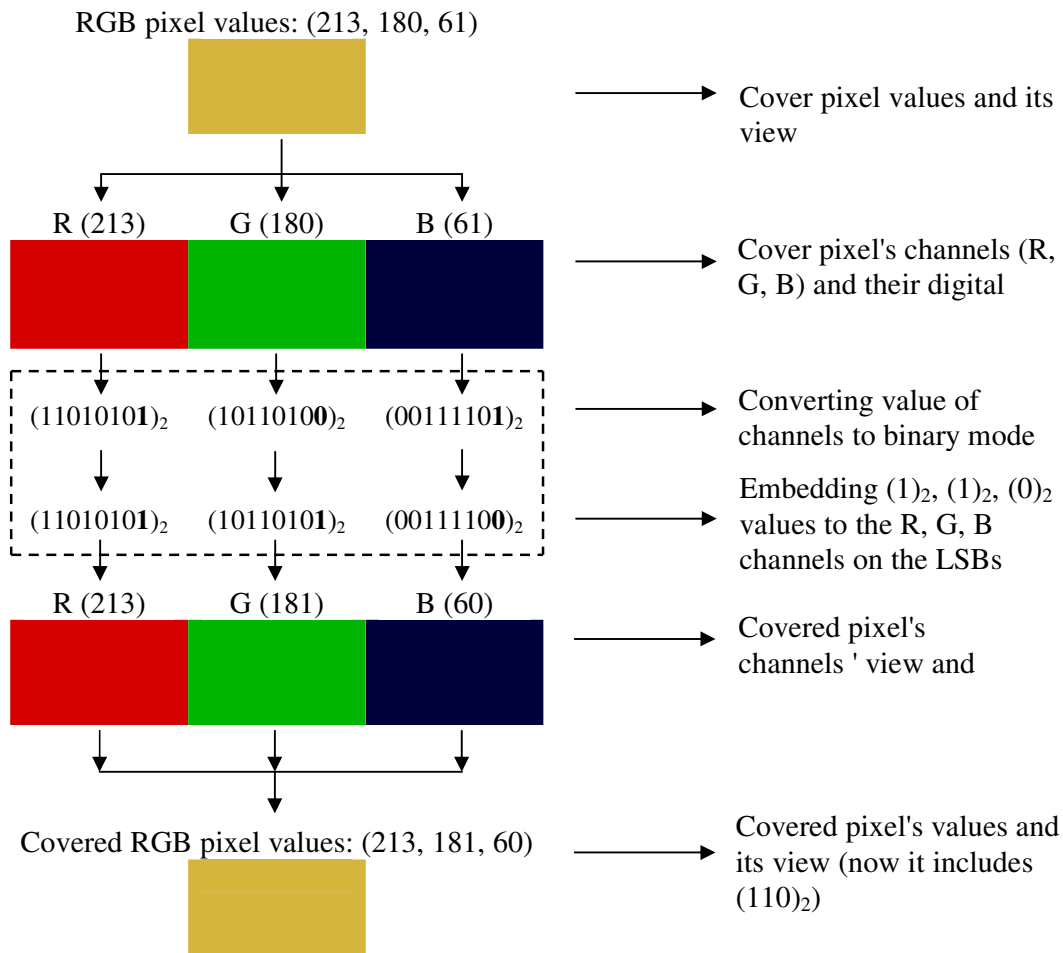


Figure 1. An example LSB data hiding example for a RGB pixel.

In a typical full-color image that is including 24 bits/pixel as seen in Figure 1. It means that 8 bits assigned to each color components. In a gray scale image, 8 bits/pixel are used. A digital image consists of a matrix of color and intensity values.

Color image files can get quite large but some compression schemes have been developed to decrease the storage and communications requirements of handling image files. Bitmap (.bmp) and GIF (.gif) files use a lossless compression algorithm. With the lossless compression algorithm, the decompressed image is identical to the original image (i.e., the image before compression). JPEG (.jpg) files use a lossy compression algorithm that approximates the image being compressed. With the lossy compression algorithm, the decompressed image is nearly the same as, but not identical to, the original image. Although data hiding can be applied on compressed images, it is more complex than data hiding on raw images (i.e., bmp, raw, etc.).

3. The Proposed Partial Optimization Method Fundamentals

There are five components needed to understand partial optimization based data hiding processes. There is a carrier, technically called "cover" denoted by the letter " c ". The letter " m " denotes the secret message that needs to be hidden. Partial optimization denoted letter " p " that cover image is separated eight distinct parts by this stage. The next is the output called covered image, denoted by the letter " s ", into which the message m needs to be embedded. Lastly, the data hiding encryption-key is denoted by " k " [10].

The output s is obtained by using " $c + m + k + p$ " into the data embedding encoder or technique. The data extraction/decoder process is to be constituted by three components. The key that is using in coding process, is necessitated in decoding

procedure. The data extraction layer takes into account regional optimizations (which are used in coding procedure) in the decoding processes (Figure 2). Otherwise it isn't attained the original message.

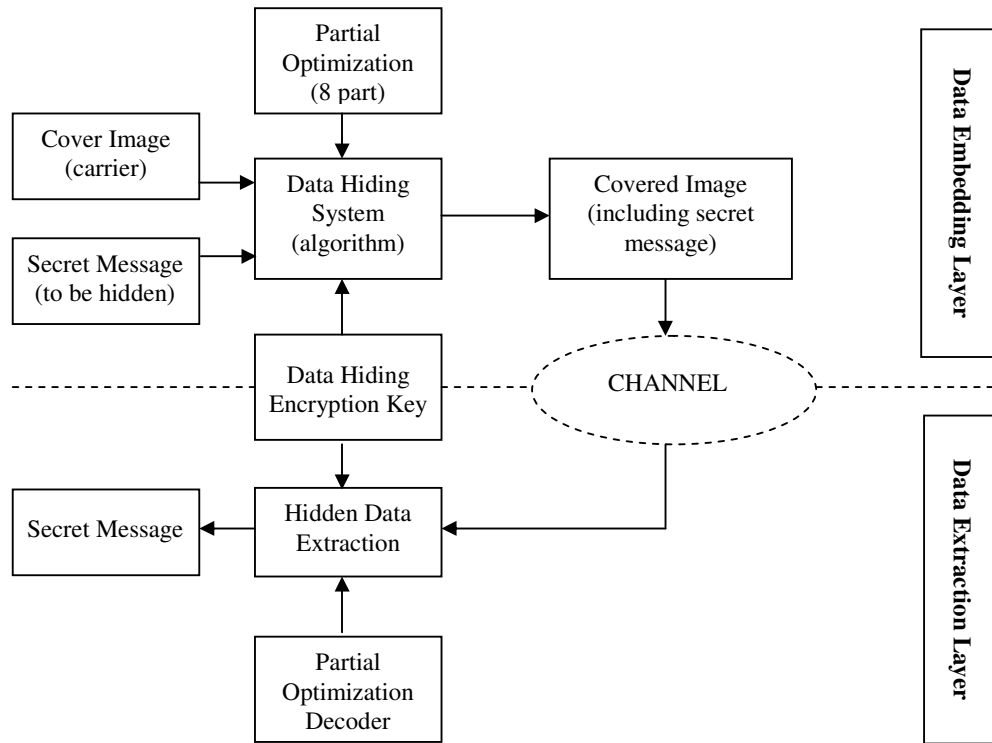


Figure 2. Overview of the proposed partial optimization based embedding/extraction approach.

Figure 3 shows that original image is separated eight portions when beginning the data hiding process. Each of the parts is to be subjected to eight distinct optimization processes.

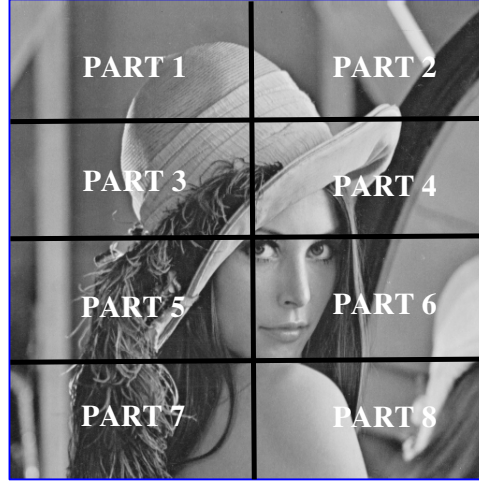


Figure 3. Original Lena image is separated eight distinct parts.

This means that embedding data is subjected to optimization with eight distinct queues while the encoding process of the first part, as is shown in Figure 4. The number of optimization can be increased to 2^8 for these queues. But spent time likewise grows. Thus, the number of optimizations is selected 2^3 in the presented work (Figure 4) [10]. It can be seen in Figure 4 that the replacement process is very easy. Embedding data is “D₇, D₆, D₅, D₄, D₃, D₂, D₁, D₀” for the first optimization but in second optimization this order is changed as “D₀, D₁, D₂, D₃, D₄, D₅, D₆, D₇”.

Optimization 1 :	D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
Optimization 2 :	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
Optimization 3 :	D ₇	D ₆	D ₅	D ₄	D ₀	D ₁	D ₂	D ₃
Optimization 4 :	D ₀	D ₁	D ₂	D ₃	D ₇	D ₆	D ₅	D ₄
Optimization 5 :	D ₇	D ₅	D ₃	D ₁	D ₆	D ₄	D ₂	D ₀
Optimization 6 :	D ₀	D ₂	D ₄	D ₆	D ₁	D ₃	D ₅	D ₇
Optimization 7 :	D ₁	D ₃	D ₅	D ₇	D ₀	D ₂	D ₄	D ₆
Optimization 8 :	D ₆	D ₄	D ₂	D ₀	D ₇	D ₅	D ₃	D ₁

Figure 4. D₀ to D₇ bits order for an ASCII embedding word.

Figure 5 illustrates all the changing states. This variations are applied to eight parts of the image. It can be easily seen that Optimization 1 is the classical LSB data hiding methods' bit array [10].

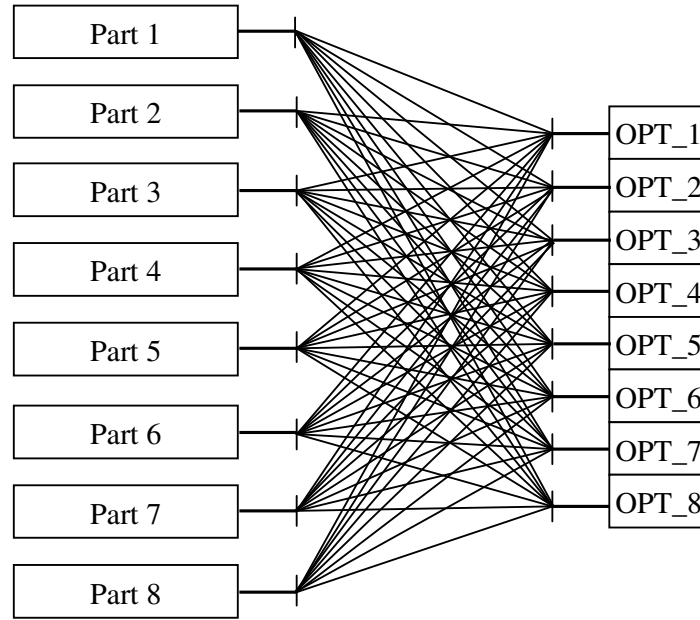


Figure 5. The process of the optimization.

The most effective outcome, which has been obtained from each region, means that the number of the altered bit is minimum. In this way the minimal values that have been attained from eight regions compose extracting that have less bit error rate. Finally, the most suitable optimization method is determined for each image part and LSB data hiding method is applied on the cover image.

4. Experimental Results

Table 1 shows that example results of the proposed optimization based LSB data hiding technique for the Lena image (512×512×8). The number of total hidden bits is 262144, it means that 32768 Bytes or 32 Kbytes. Thus it can be easily said that the bit rate is 1

bpp. Considering the results obtained from the software developed for this presented approach, it seems impossible for the LSB coding technique to have the less distortion than our method because only a unique method is implemented to all regions of the cover image. However, when eight different optimizations are implemented to eight different regions, the most optimum method can be easily determined for each region and the best outcome can be obtained for the cover image.

Table 1. All optimization results for all image parts (512×512×8 Lena image).

		Opt_1	Opt_2	Opt_3	Opt_4	Opt_5	Opt_6	Opt_8	Opt_7
PART 1	A	16257	16191	16321	16167	16139	16219	16241	16235
	B	16503	16569	16439	16593	16621	16542	16519	16525
	C	50.37	50.57	50.18	50.65	50.73	50.37	50.42	50.44
PART 2	A	16379	16483	16297	16497	16423	16525	16413	16417
	B	16381	16277	16463	16263	16337	16232	16347	16343
	C	50.00	49.68	50.25	49.64	49.86	49.55	49.89	49.88
PART 3	A	16407	16391	16419	16467	16397	16469	19479	16419
	B	16353	16369	16341	16293	16363	16291	16281	16341
	C	49.91	49.96	49.88	49.73	49.94	49.72	49.69	49.88
PART 4	A	16502	16464	16504	16430	16418	16452	16460	16508
	B	16258	16296	16256	16330	16342	16308	16300	16252
	C	49.62	49.74	49.62	49.84	49.88	49.78	49.75	49.60
PART 5	A	16595	16499	16587	16527	16527	16535	16473	16573
	B	16165	16261	16173	16233	16233	16225	16287	16187
	C	49.34	49.63	49.36	49.55	49.55	49.52	49.71	49.41
PART 6	A	16391	16377	16383	16381	16377	16371	16365	16395
	B	16369	16383	16377	16379	16383	16389	16395	16365
	C	49.96	50.00	49.99	49.99	50.00	50.02	50.04	49.95
PART 7	A	16613	16605	16575	16591	16549	16597	16599	16627
	B	16147	16155	16185	16169	16211	16163	16161	16133
	C	49.28	49.31	49.40	49.35	49.48	49.33	49.33	49.24
PART 8	A	16378	16336	16370	16348	16400	16370	16352	16360
	B	16446	16488	16454	16476	16424	16454	16472	16464
	C	50.20	50.32	50.22	50.29	50.13	50.22	50.28	50.25

A, B and C as seen in Table 1 denote the numbers of different bits for related part, the numbers of same bits for related part and bit rate for same bits to the number of all

LSBs for related part, respectively. Opt_X denotes that Optimization_X. Considering the results shown in Table 1, they can be easily listed in the following statistics (Table 2).

Table 2. The number of prevented pixel distortions for the test image Lena.

Part Number	The number of changed LSB for classical LSB data hiding	The number of changed LSB for proposed optimization based data hiding	The number of prevented pixel distortions
Part 1	16257	16139	118
Part 2	16379	16297	82
Part 3	16407	16391	16
Part 4	16502	16418	84
Part 5	16595	16473	122
Part 6	16391	16365	26
Part 7	16613	16549	64
Part 8	16378	16336	42
The number of total prevented pixel distortion			554

Classical LSB embedding technique uses the first order of optimization method (Opt_1) and it has 131522 error bits for this sample (please see the numbers on the first column, A values in Table 1). However, proposed method provides less error bits employing distinct optimization regions. The number of total prevented pixel distortion is 554 bits for this example because the number of error (distorted) bits are equal to 130968 for the proposed method. It is determined that most suitable alternative which has least error bits rate in this method. This rule is operated each region of the eight parts for the image. Thus, it is acquired less error bits rate for each region and the all areas. For example fifth optimization (Opt_5) is selected for the Part 1 and the third optimization (Opt_3) is Part 2 for this example application. On the other hand, Opt_2, Opt_5, Opt_7, Opt_7, Opt_5, Opt_2 are selected for Part 3, Part 4, Part 5, Part 6, Part 7, Part 8, respectively.

Figure 6 shows that the original Lena image (a) and coded by the classical LSB method (b) and the proposed optimization based (LSB_{OPT}) method (c).



(a)



(b)



(c)

Figure 6. The comparison of the perceptual results the original (a), coded by classical LSB method (b) and coded by the proposed method (c).

Figure 7 shows that other reference images (Baboon, Barbara, F16, Goldhill, Houses and Peppers) used for experimental applications.

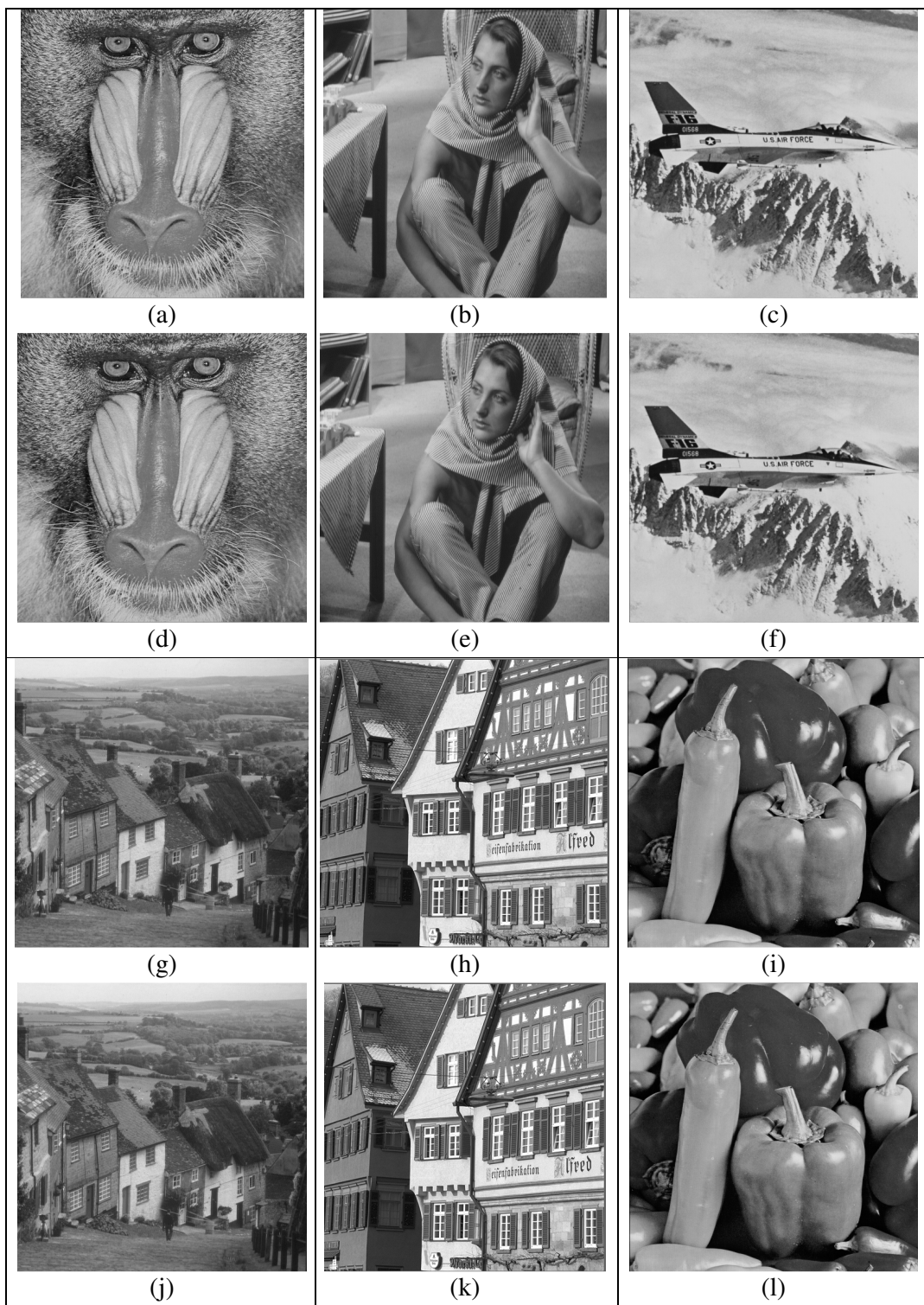


Figure 7. The $512 \times 512 \times 8$ reference images (Baboon (a), Barbara (b), F16 (c), Goldhill (g), Houses (h) and Peppers (i)) and their covered versions (hidden data: 32 KB) encoded by the proposed method.

After the experimental applications, we have obtained the numbers of total prevented pixel distortion for all test images (Table 3). The results show that the proposed method causes less distortion effect than the classical LSB method on the image.

Table 3. The number of total prevented bits for each test image considering the embedded data.

Embedded Data		2 KB	4 KB	8 KB	16 KB	32 KB
Lena	The number of total prevented bits	190	218	320	412	554
Baboon		184	202	290	326	360
Barbara		258	262	380	522	886
F16		232	316	484	570	666
GoldHill		162	262	386	440	742
Houses		192	292	344	430	594
Peppers		224	232	434	560	726

Figure 8 shows that the number of total prevented bits graphically for each test image considering the embedded data.

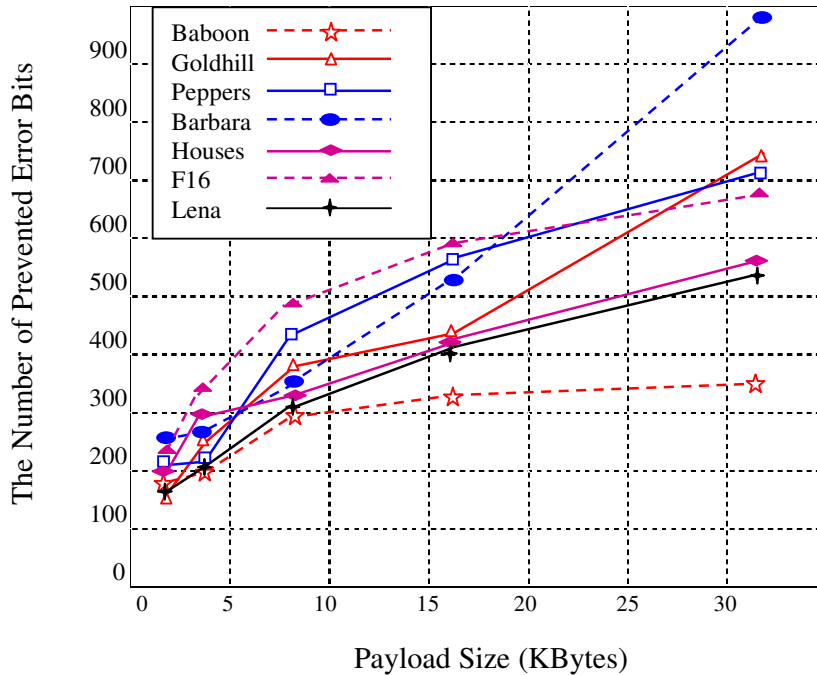


Figure 8. The numbers of prevented error bits for each test images.

In order to measure the image quality, Mean Square Error (MSE) and Peak Signal to Noise Ratio (PSNR) are usually used in the literature. The MSE should be computed first as given in Equation (1) [11], [12] then the PSNR can be derived as in Eq. (2) [13], [14], [15]. Where “ O ” and “ C ” are the original image and the covered image pixel values (binary) respectively to be compared and the image size is “ $m \times n$ ”. Note that, Eq. (1) is specified for only monochrome images; for color images, the denominator of the Eq. (2) is multiplied by a factor 3.

$$MSE = \frac{1}{m \times n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \|O(i, j) - C(i, j)\|^2 \quad (1)$$

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (2)$$

Here, MAX is the maximum BV of the pixels in an image. MAX can be between 0-255 when pixels are presented by 8 bits. The PSNR quality metric has been used for comparing the classical LSB data hiding method and the proposed method (LSB_{OPT}). Considering the PSNR results, Table 4 shows that the proposed method works better than the classical LSB data hiding method.

Table 4. The PSNR comparison of the classical LSB and the proposed LSB_{OPT}.

Message Size	32 KB		16 KB		8 KB		4 KB		2 KB	
Data Hiding Method	LSB	LSB _{OPT}	LSB	LSB _{OPT}	LSB	LSB _{OPT}	LSB	LSB _{OPT}	LSB	LSB _{OPT}
Lena	51.14	51.16	54.14	54.17	57.16	57.20	60.19	60.25	63.17	63.27
Baboon	51.14	51.16	54.13	54.15	57.16	57.20	60.16	60.22	63.13	63.22
Barbara	51.13	51.16	54.14	54.18	57.14	57.20	60.21	60.28	63.09	63.23
F16	51.15	51.17	54.15	54.19	57.15	57.22	60.16	60.24	63.17	63.29
Goldhill	51.14	51.17	54.14	54.17	57.17	57.22	60.18	60.25	63.11	63.20
Houses	51.21	51.23	54.18	54.21	57.16	57.21	60.19	60.27	63.16	63.26
Peppers	51.15	51.18	54.14	54.18	57.15	57.21	60.16	60.22	63.22	63.34
Average Improvement	0.24		0.32		0.52		0.68		1.08	

Considering the above valuable results, the proposed partial optimization based LSB data hiding is more efficient than the classical LSB data hiding technique because LSB_{OPT} causes relatively less distortion on the cover images while same has same bit rate values. The PSNR performance of the LSB_{OPT} is better than the classical LSB varying between 0.24 dB and 1.08 dB as seen in Table 4.

The proposed LSB_{OPT} has been evaluated in terms of other quality metrics for perceptual distortion. The PSNR-Human Visual System (PSNR-HVS) and the PSNR-HVS-Modified (PSNR-HVS-M) are modified versions of the classical PSNR method, intended to take into account the effect of the HVS Contrast Sensitivity Function (CSF) and contrast masking. The PSNR-HVS is based on the PSNR and the Universal Image Quality Index (UQI) [16] modified to take into account the HVS properties (Figure 9) [17].

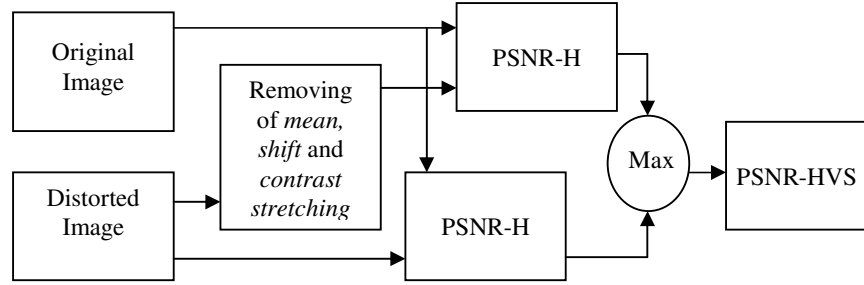


Figure 9. PSNR-HVS calculation [17].

The PSNR-HVS performance of the LSB_{OPT} is better than the classical LSB varying between 1.49 dB and 1.58 dB as seen in Table 5.

Table 5. The PSNR-HVS comparison of the classical LSB and the proposed LSB_{OPT} .

Message Size	32 KB		16 KB		8 KB		4 KB		2 KB	
Data Hiding Method	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}
Lena	47.02	48.61	50.03	51.61	53.05	54.63	56.09	57.66	59.10	60.62
Baboon	47.01	48.60	50.02	51.58	53.05	54.62	56.06	57.62	59.08	60.57
Barbara	47.03	48.58	50.03	51.60	53.04	54.59	56.10	57.70	59.08	60.49
F16	47.02	48.61	50.04	51.61	53.06	54.61	56.08	57.63	59.12	60.63
Goldhill	47.02	48.61	50.03	51.59	53.07	54.63	56.09	57.64	59.06	60.55
Houses	47.07	48.67	50.04	51.64	53.06	54.62	56.11	57.66	59.10	60.61
Peppers	47.02	48.62	50.04	51.61	53.06	54.61	56.07	57.62	59.14	60.67
Average Improvement	1.58		1.57		1.56		1.56		1.49	

The PSNR-HVS-M is a simple and effective model of visual between-coefficient contrast masking of DCT basis functions based on the HVS. The model operates with the values of DCT coefficients of 8×8 pixel block of an image. For each DCT coefficient of the block the model allows to calculate its maximal distortion that is not visible due to the between-coefficient masking. A modification of the PSNR is also

described in this paper. The PSNR-HVS-M takes into account the contrast sensitivity function (CSF) (Figure 10) [18].

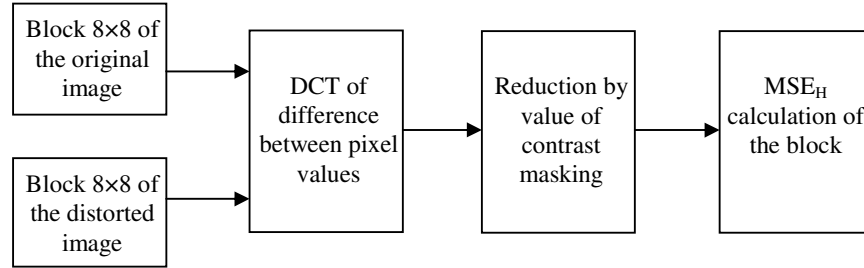


Figure 10. PSNR-HVS-M calculation [18].

The PSNR-HVS-M performance of the LSB_{OPT} is better than the classical LSB varying between 2.71 dB and 2.86 dB as seen in Table 6.

Table 6. The PSNR-HVS-M comparison of the classical LSB and the proposed LSB_{OPT} .

Message Size	32 KB		16 KB		8 KB		4 KB		2 KB	
Data Hiding Method	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}	LSB	LSB_{OPT}
Lena	47.04	49.91	50.06	52.91	53.11	55.93	56.20	58.98	59.22	61.93
Baboon	47.04	49.96	50.03	52.92	53.11	55.98	56.12	58.96	59.13	61.88
Barbara	47.04	49.89	50.06	52.90	53.09	55.90	56.23	59.04	59.14	61.79
F16	47.05	49.89	50.07	52.89	53.12	55.87	56.15	58.87	59.23	61.90
Goldhill	47.06	49.94	50.05	52.91	53.15	55.97	56.20	58.98	59.09	61.86
Houses	47.22	50.06	50.11	52.99	53.13	55.95	56.27	59.03	59.20	61.95
Peppers	47.06	49.94	50.07	52.90	53.12	55.93	56.14	58.94	59.34	62.05
Average Improvement	2.86		2.85		2.81		2.79		2.71	

5. Conclusions

In this presented work, a new partial optimization method (LSB_{OPT}) for LSB data hiding in digital images is introduced. At the first step of the LSB_{OPT} , the original cover

image is separated into eight portions before beginning the data hiding process. Then, each part is to be subjected to the eight distinct optimization processes. This means that embedding data is subjected to optimization with eight distinct queues while the encoding process of the first part. Using the proposed approach, the most effective outcome obtained from the eight different regions ensures that the number of the altered bits on the cover image is minimized. Finally, the most suitable optimization method is determined for each image part and then the LSB data hiding method is applied on the cover image.

Experimental results show that not only the proposed method's PSNR performance but also PSNR-HVS and PSNR-HVS-M performance are better than the classical LSB data hiding method. The performance of the LSB_{OPT} can be easily increased by multiplying the number of optimization.

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